Bureau
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Annexe 2
Annex 2

Valeurs recommandées des fréquences étalons destinées à la mise en pratique de la définition du mètre et aux représentations secondaires de la seconde

Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second

Partie 1
Part 1

Iodine ()

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CCL-CCTF Frequency Standards Working Group

May 2016
mai 2016

Bureau International
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Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second
Part 1: Iodine ()

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1. Recommended value [1]

$f\left({}^{127}\mathrm{I}_2\right) = 564\ 074\ 632.42\ \mathrm{MHz}$

equivalent to

$\lambda = 531\ 476\ 582.65\ \mathrm{fm}$

with an estimated relative standard uncertainty of 1×10^{-10} applies to the radiation of a frequency-doubled diode DFB laser, stabilized with an iodine cell external to the laser with the following parameters:

- cold-finger temperature (25 ± 0.5) °C (corresponding to the iodine pressure $p = 41$ Pa)
- frequency modulation width, peak-to-peak, (12 ± 1) MHz for $3f$ detection cases;
- saturating beam intensity of $12.7\ \mathrm{mW}\ \ (\pm 10\%)$ at a beam diameter of $1\ \mathrm{mm}$

2. Source data

Adopted value $f = 564\ 074\ 632.42$ MHz
$u_c/y = 1 \times 10^{-10}$

taken from:

f/ kHz	u_c/y	source data
564 074 632 419 (8)	1.4×10^{-11}	[2]

This value was issued from only a single laboratory under conditions where strong linear absorption at Doppler centre occurs, thereby degrading the signal-to-noise level at this high iodine vapour pressure [2]. Thus, the CIPM following a recommendation of the CCL considered it prudent to enlarge the standard uncertainty by a factor of seven, and round to 1×10^{-10}.

Absorbing atom $^{127}I_2$, a_1 component, R(36) 32-0 transition

References

[1] CIPM Recommendation 2 (CI-2015): Updates to the list of standard frequencies http://www.bipm.org/jsp/en/CIPMRecommendations.jsp

[2] T. Kobayashi, D. Akamatsu, K. Hosaka, H. Inaba, S. Okubo, T. Tanabe, M. Yasuda, A. Onae, F.-L. Hong, Compact iodine-stabilized laser operating at 531 nm with stability at the 10^{-12} level and using a coin-sized laser module, *Optics Express*, **23**, 20749 (2015).